

Lecture 4

- Forms
- <input>-element
 - type-attribute
 - other attributes
- Other form elements
- Putting form together
- GET and POST
- Form validation
 - Pattern®ex

Forms

- Forms can be used in the implementation of
 - e.g., web stores, search engines, information services (searching from libraries, databases), polls, registration sites, etc.
 - the information is passed on to a server (HTTP transaction)
 - can also be consumed by the web page itself
- A form contains one or more user inputs, like
 - text fields, checkboxes, radio buttons, select boxes, etc.

Suitable elements for forms

- Input elements (content category: interactive)
 - `<input>`, `<button>`, `<datalist>`, `<option>`, `<optgroup>`, `<textarea>`, `<keygen>`, `<output>`, `<progress>`, `<meter>`
 - `<form>`, `<label>`, `<fieldset>`, `<legend>` for **organizing the elements**
- Most versatile element is the `<input>`-element

<input> types

```
<label for="idx">label for the field</label>  
<input type="text" id="idx">
```



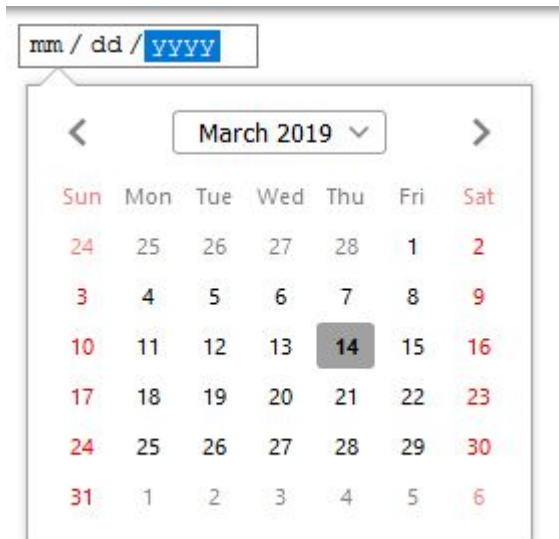
- Type attribute defines how the element is displayed and how it behaves
- The default value of `type` is `text`.
- Other types include `number` and more elaborate types, such as:
 - `password` – hides the given input automatically
 - `email` – for getting an email address
 - `tel` – for getting a phone number

Yet more `<input>` types

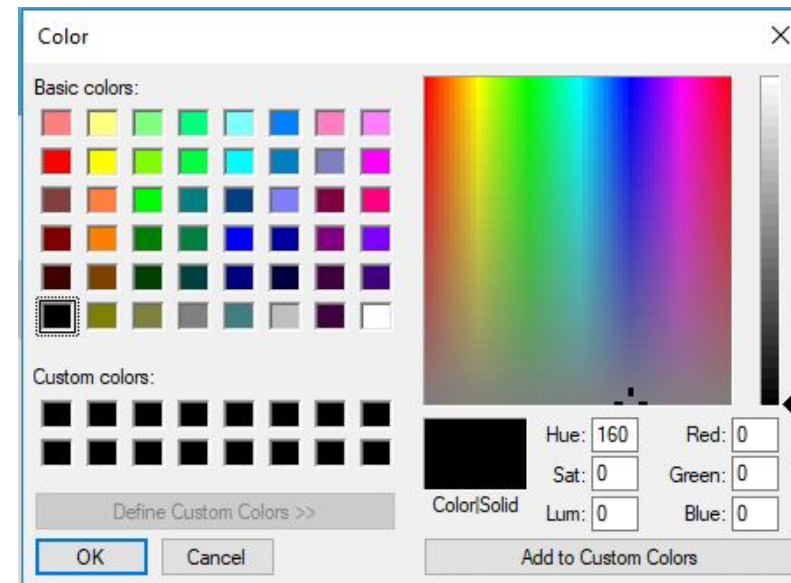
- `radio` – radio buttons (grouping on the basis of `name` attribute)
- `checkbox` – allows choosing several options (also by `name` attribute)
- `button` – button element with text inside, no default behaviour
- `file` – for choosing a file from the computer
- `image` – to implement a graphical button (using an image)
- `hidden` – not displayed but its value is submitted to the server

New HTML5 input types are handy

`<input type="date">`



`<input type="color">`



Wufoo: <input> element list

<https://www.wufoo.com/html5>

- How different browsers support different `type` attributes of the `<input>`-element

								
		Firefox	Safari	Safari	Chrome	Opera	IE	Android
Email	?	4+	5+	3.1+	6+/10+	10.6+	10+	4+
Tel	?	4+	5+	3.1+	6+	10.6+	10+	2.3+
Url	?	4+	5+	3.1+	6+/10+	10.6+	10+	2.3+
Search	?	4+	5+	4+	6+	10.6+	9/10+	4+
Color	?	29+	8+	8-	20+	11+	11-	4.4+
Number	?	29+	5+	3.2+	7+	9+	10+	2.3+
Range	?	23+	4+	5+	6+	11+	10+	4.2+
Date	?	32-	7-	5+	20+	9+	11-	4.4+
Text	?	All	All	All	All	All	All	All
Button	?	All	All	All	All	All	All	All
Hidden	?	All	All	All	All	All	All	All
Password	?	All	All	All	All	All	All	All
Checkbox	?	All	All	All	All	All	All	All
Radio	?	All	All	All	All	All	All	All
File	?	All	All	All	All	All	All	All
Submit	?	All	All	All	All	All	All	All
Reset	?	All	All	All	All	All	All	All

Input element, other attributes

- Browser support for other attributes of the `<input>`-element
- Scan these through with Wufoo live demos
- Browser ignores no-support features

	 Firefox	 Safari	 Safari	 Chrome	 Opera	 IE	 Android
Placeholder	4+	4+	4+	10+	11.10+	10+	2.3+
Autofocus	4+	5+	5-	6+	11+	10+	3+
Maxlength	4.4+	5+	4+	6+	11+	9+/10	2.3+
List (Datalist)	4+	7-	7-	20+	9+	10+	4.3-
Autocomplete	4+	5.2+	6+	14+	10.6+	11+	4.4+
Required	6+	5+	4+	6+	10.6+	10+	2.3+
Pattern	4+	5+	4+	10+	11+	10+	2.3+
Spellcheck	3.6+	4+	7+	10+	11+	10+	4.3-
Novalidate	4+	7-	7-	10+	10+	10+	4.2-
Formnovalidate	4+	7-	7-	6+	10.6+	10+	4.2-
Formaction	4+	5.2+	5+	10+	10.6+	10+	4.0+
Formmethod	4+	5.2+	5+	10+	10.6+	10+	4.0+
Formtarget	4+	5.2+	5+	10+	10.6+	10+	4.0+

Formenctype	4+	5.2+	5+	10+	10.6+	10+	4.0+
minlength	28-	7-	7-	33-	20-	11-	4.4-
readonly	All	All	All	All	All	All	All
disabled	All	All	All	All	All	All	All
size	All	All	All	All	All	All	All
name	All	All	All	All	All	All	All
height, width & source	All	All	All	All	All	All	All
checked	All	All	All	All	All	All	All
tabindex	All	All	All	All	All	All	All
form	4+	5.1+	5+	10+	9.5+	11+	4.0+
inputmode	FF OS	7-	7-	34-	20-	10-	4.4-
dirname	29-	6+	6+	16+	10.6+	11-	4.4+
Accept	11+	5.2+	7+	10+	10.6+	10+	4.2-
Multiple	3.6+	5+	7+	6+	11+	10+	4.7+
Min / Max / Step	16+	5+	3.2	6+	10.6	10+	4+

Besides input, other elements exist

text area		<code><textarea></code>	– multi-line text field
select		<code><select></code>	– drop-down widget
multiple choice		<code><select multiple></code>	– multiple values from a list can be selected
datalist		<code><datalist></code>	– search from a list is autocompleted
		<code><button></code>	– type = submit reset

HTML5 elements and browser support

							
	Firefox	Safari	Safari	Chrome	Opera	IE	Android
Meter	16+	5.2+	8-	8+	11+	11-	4.4+
Progress	6+	5.2+	7+	6+	10.6+	10+	4.4+
Output	4+	5.1+	5+	10+	11+	10+	2.3+
Keygen	3.6+	4+	7-	6+	10.6+	11-	2.3+
Datalist	4+	8-	8-	20+	9+	10+	4.4+
Option	All	All	All	All	All	All	All
Label	All	All	All	All	All	All	All



Putting it together

- `<form>` contains multiple interactive elements, mainly inputs
- Submit button
 - sends them to a specified form handler
 - as name-value pairs

Lomake

* Käyttäjänimi: * Salasana:
`<input type="text">` ja `<input type="password">`

Sähköpostiosoite: `<input type="email">`

Puhelin: `<input type="tel">`

Ikäsi: * `<select>`

Missä kuussa olet syntynyt: `<datalist>`

Interaktiivisia elementtejä voi ryhmitellä `<-fiedset>` ja `<legend>`-elementtien avulla

Mistä musiikista pidät eniten:

- ☒ Rock
- ☐ Pop
- ☐ Jazz
- ☐ Kansanmusiikki
- ☐ Klassinen

`<input type="radio">`

Millä sitä kuuntelet? ☐ puhelimella ☐ tietokoneella ☐ cd-soittimella ☐ radiolla
`<input type="checkbox">`

Millä näistä pääasiassa?

`<input type="text" value="puhelimella">`

Varmaan osaat soittaa triangelia, mutta soitatko joitain muuta seuraavista instrumenteista (voit valita useampia pitämällä control-painiketta alhaalla, Macillä command-painiketta)?

kitara
rummut
piano
basso

Mikä on mielipiteesi esitetystä asiasta?

* Pakollinen kenttä.

Constructing a form

```
<form action="form-handler" method="get"|"post">  
  [input widget]*  
  {submit button}  
</form>
```

- **Attribute** `action` specifies a form handler
 - the form handler is often given as a URL
 - runs on a separate server
 - written for example with php, perl, ...
 - server side programming not part of this course (but e.g. WebDev1 uses node.js for the purpose)
 - **Attribute** `method` specifies how the data is sent

GET method

```
<form action= "form-handler url" method="GET">
```

- Browser sends a request to a server
 - the form data is attached to the address
 - query string: ?key=value[&key=value]*
 - key is connected to a value with equal sign (=)
 - pairs are separated with 'and' (&)
 - GET used, if visibility of values is not an issue
 - search engines often use the GET method
 - gives a possibility to save the query and send it to other users
 - queries can be sent even without the form

POST method

```
<form action="server url" method="POST">
```

Browser sends a request to a server

- the input data is sent in the body of HTTP request
- the data is hidden from the user and other onlookers
- however
 - also with the POST method the data is open (not encrypted)
 - a third party is capable of reading the sent data (*also the password*)
- sensitive data between must be sent with SSL-encrypted HTTPS protocol

JWT

0151

JWT

0152

Form validation

- Before submission, the data should be checked
 - general validations specified with elements or attributes are performed before sending
 - e.g. mail-address format, required fields, min-lengths of input values, etc.
- The form is submitted with a submit button

Input pattern by regular expressions

- In addition, values can be given specifically defined patterns using regular expressions (regex)
 - patterns that can be used to match character combinations in text strings

Regular expressions

- a — matches one character that is a (not b, not aa, etc.)
- abc — matches a, followed by b, followed by c
- a* — matches the character a, zero or more times (+ matches a character one or more times)
- ^a — matches a string which starts with a
- a|b — matches one character that is a or b
- [abc] — matches one character that is a, b, or c
- ^[abc] — matches a string starting with a, b, or c (note the difference with [^abc])
- [a-z] — matches any character in the range a–z, lower case only ([A-Za-z] for lower and upper case, and [A-Z] for upper case only)
- a.c — matches a, followed by any character, followed by c
- a{5} — matches a, 5 times
- a{5,7} — matches a, 5 to 7 times, but no less or more

Numbers and other characters in these expressions too, such as:

- [-] — matches a space or a dash
- [0-9] — matches any number in the range 0 to 9

Examples of regular expressions

`[Mm].*k` — A single character that is an upper or lowercase m, followed by zero or more characters of any type, followed by a single lowercase k.

`[A-Z][a-z]+` — A single upper case character followed by one or more lower case letters.

`[0-9]{3}[-][0-9]{7}` — A simple match for a Finnish domestic phone number — three numbers, followed by a space or a dash, followed by seven numbers

```
<input type="text" id="town" pattern="[A-Z][A-Za-z' -]+" minlength="2">
```

Accepts Manchester, Ashton-under-lyne, and Bishop's Stortford.